

Program 4: Maven Project To Connect To Web Servers Using Selenium Dependency (PracticeAutomation.com)

Create a Maven Project:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```

rakshith@rakshith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp02 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom -----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/rakshith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp02
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/rakshith/devops/MyMavenSeleniumApp02
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.641 s
[INFO] Finished at: 2026-03-19T14:24:44+05:30
[INFO]

```

Open pom.xml and add all the required dependencies:

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenPracticeAutomationApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenPracticeAutomationApp</name>
  <url>http://maven.apache.org</url>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>

    <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
    <dependency>
      <groupId>org.seleniumhq.selenium</groupId>
      <artifactId>selenium-java</artifactId>
      <version>4.29.0</version>
    </dependency>
  </dependencies>

```

```

<!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
</plugins>
</build>
</project>

```

```

</plugin>

```

```

</plugins>
</build>
</project>

```

Edit App.java:

```

App.java
~/Devops/MyMavenSeleniumApp02/src/main/java/com/example

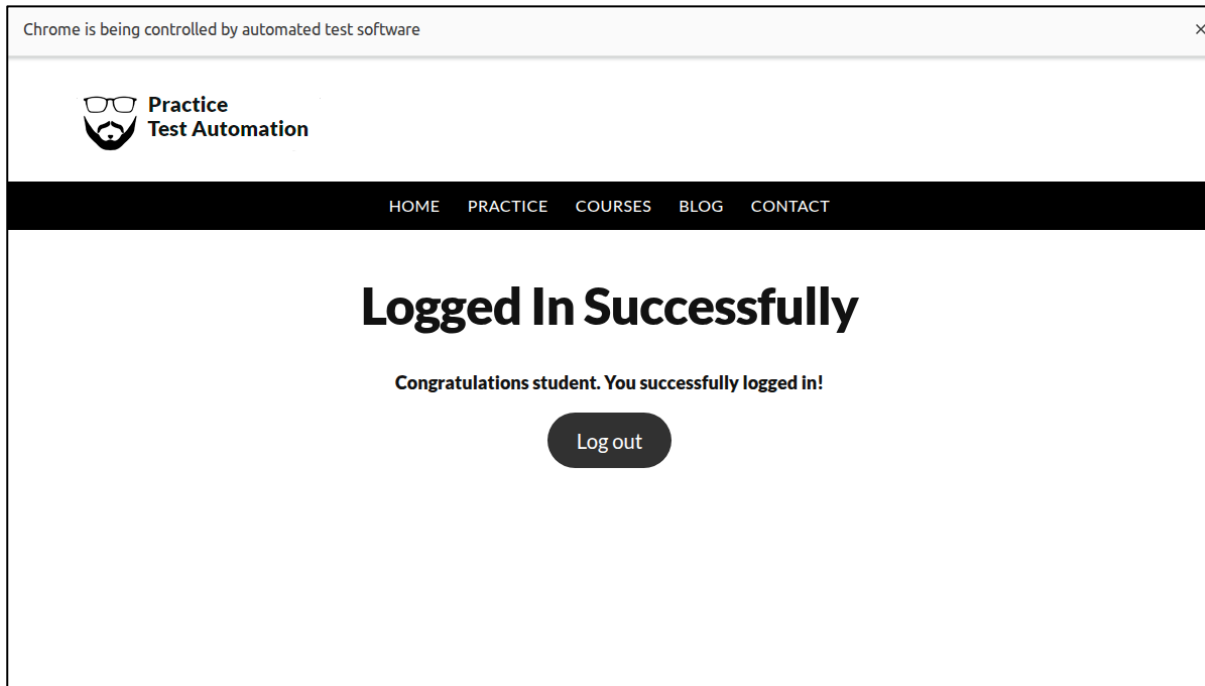
1 package com.example;
2 import org.openqa.selenium.By;
3 import org.openqa.selenium.WebDriver;
4 import org.openqa.selenium.chrome.ChromeDriver;
5 public class App
6 {
7     public static void main( String[] args )
8     {
9         WebDriver driver=new ChromeDriver();
10        driver.get("https://www.practicetestautomation.com/practice-test-login/");
11        driver.manage().window().maximize();
12        driver.findElement(By.id("username")).sendKeys("student");
13        driver.findElement(By.id("password")).sendKeys("Password123");
14        driver.findElement(By.id("submit")).click();
15    }
16 }

```

Build the project using the command: mvn clean install

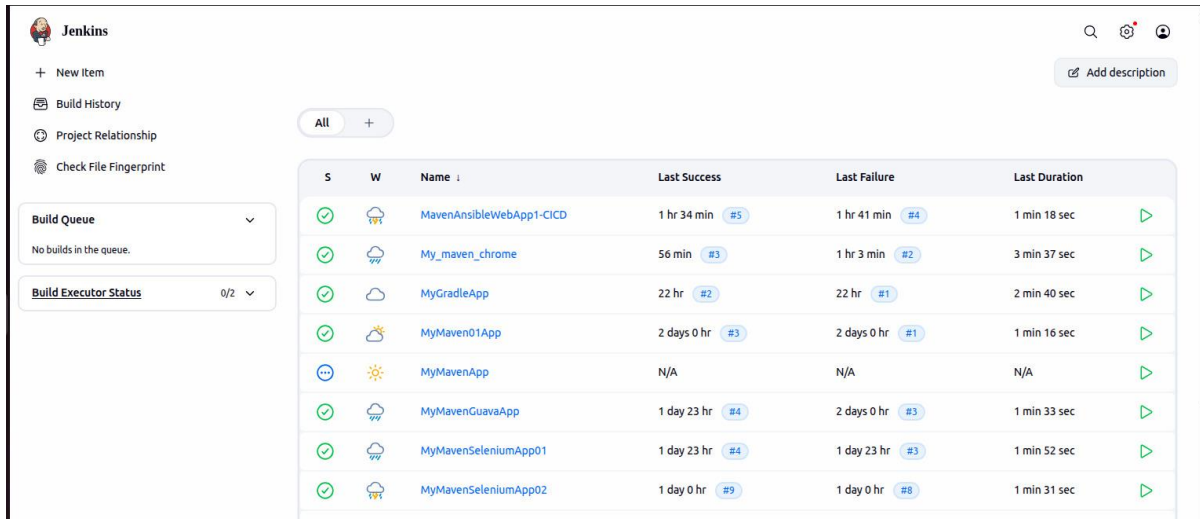
Now Run the command using

mvn exec:java -Dexec.mainClass="com.example.App"



Jenkins Pipeline for MyMavenSeleniumApp02:

Step 1:Open Jenkins Dashboard to create a new item



The screenshot shows the Jenkins Dashboard. On the left, there is a sidebar with navigation links: '+ New Item', 'Build History', 'Project Relationship', and 'Check File Fingerprint'. Below these are two dropdown menus: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2'). The main area displays a table of build items. The table has columns for 'S' (Status), 'W' (Icon), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. The items listed are:

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	MavenAnsibleWebApp1-CICD	1 hr 34 min #5	1 hr 41 min #4	1 min 18 sec
✓	☁	My_maven_chrome	56 min #3	1 hr 3 min #2	3 min 37 sec
✓	☁	MyGradleApp	22 hr #2	22 hr #1	2 min 40 sec
✓	☁	MyMaven01App	2 days 0 hr #3	2 days 0 hr #1	1 min 16 sec
...	☀	MyMavenApp	N/A	N/A	N/A
✓	☁	MyMavenGuavaApp	1 day 23 hr #4	2 days 0 hr #3	1 min 33 sec
✓	☁	MyMavenSeleniumApp01	1 day 23 hr #4	1 day 23 hr #3	1 min 52 sec
✓	☁	MyMavenSeleniumApp02	1 day 0 hr #9	1 day 0 hr #8	1 min 31 sec

Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenSeleniumApp02” and Item type as Pipeline and Create it

New Item

Enter an item name

MyMavenSeleniumApp02

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

OK

Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/rakshith31-hub/MyMavenSeleniumApp02.git

Credentials ?

GitHub Credential

+ Add

Advanced

Save Apply

Step 4: Now in Terminal, go to directory created for MyMavenSeleniumApp02 and create a new file called “Jenkinsfile”

Paste the code inside Jenkinsfile

Enter following code into jenkins file

```
pipeline
{
    agent any // Use any available agent

    tools {
        maven 'Maven' // Ensure this matches the name configured in Jenkins
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url:
'https://github.com/rakshith31-
hub/MymavenSeleniumApp02.git'
            }
        }
    }
}
```

```
}
```

```
stage('Build') { steps {  
    sh 'mvn clean package' // Run Maven build  
}
```

```
}
```

```
stage('Test') { steps {  
    sh 'mvn test' // Run unit tests  
}
```

```
}
```

```
stage('Run Application') { steps {  
    // Start the JAR application  
    sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'  
}
```

```
}
```

```
}
```

```
post {  
    success {  
        echo 'Build and deployment successful!'  
    }
```

```
    failure {  
        echo 'Build failed!'  
    }
```

```
}
```

```
}
```

Step 5: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 6: Now we need to Push changes and commit by pushing the necessary “Jenkinsfile” to the respective Repo

The screenshot shows the GitHub interface for a repository named 'MyMavenSeleniumApp02' by user 'rakshith31-hub'. The repository is public and has 4 commits. The file list shows 'src', 'target', 'Jenkinsfile', and 'pom.xml'. The 'Jenkinsfile' and 'pom.xml' files were updated yesterday. The 'README' section is currently empty, with a prompt to 'Add a README'. The right sidebar shows 'About' (no description), 'Activity' (0 stars, 0 watching, 0 forks), 'Releases' (no releases published), and 'Packages' (no packages published).

Step 7: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.

Jenkins / All / MyMavenSeleniumApp02

MyMavenSeleniumApp02 ✓ [Add description](#)

Stage View

Average stage times:
(full run time: ~1min 31s)

	Declarative: Checkout SCM	Checkout	Verify Tools	Build	Test	Declarative: Post Actions
Apr 14 13:41 1 commit	2s	2s	7s	39s	28s	497ms

Permalinks

- [Last build \(#9\), 1 day 4 hr ago](#)
- [Last stable build \(#9\), 1 day 4 hr ago](#)
- [Last successful build \(#9\), 1 day 4 hr ago](#)
- [Last failed build \(#8\), 1 day 4 hr ago](#)
- [Last unsuccessful build \(#8\), 1 day 4 hr ago](#)
- [Last completed build \(#9\), 1 day 4 hr ago](#)

Builds

14 April 2026

- ✓ #9 13:41
- ✗ #8 13:37
- ✗ #7 13:33
- ✗ #6 12:21
- ✗ #5 12:21
- ✗ #4 12:21